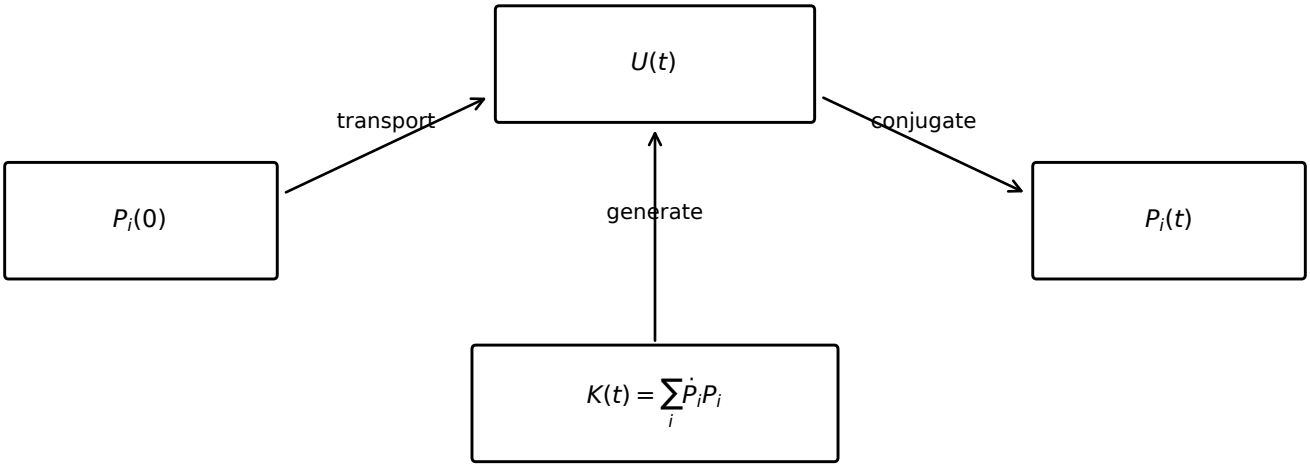Covariant transport separates physical evolution from chart motion